

Tommy You

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EDUCATION

University of Toronto

Sep 2023 – present

Bachelor of Applied Science in Mechanical Engineering + PEY Co-op — CGPA 3.76/4.00

Toronto, ON

- Relevant Courses: **Engineering biology**, Fluid Mechanics, Solid Mechanics, Material Sciences, Manufacturing Engineering, Circuits (LTspice), Thermodynamics, Kinematics and Dynamics of Machinery

WORK EXPERIENCE

Engineering Co-op

May 2026 – Present

Micromem

Toronto, ON

- Analyze laboratory sensor operating principles and design **front-end circuits**, refining **PCB schematics** for sensor platforms.
- Develop **PCB layouts and routing** and contribute to **embedded-system development** for sensor hardware.
- Support sensor development in collaboration with the **Canadian government and DRDC**, targeting defence applications and commercialization in other sensor markets.

RESEARCH EXPERIENCE

Biological Research Assistant

May 2025 – Present

Integrative Biology and Microengineered Technologies Laboratory, University of Toronto

Toronto, ON

- Performed mammalian **cell culture** under BSCs, preparing reagents and media and conducting seeding, passaging, and viability assays to ensure consistent and reliable experimental results.
- Designed and assisted in the fabrication of **microfluidic** platforms using **Solidworks** including channel layout, PDMS casting, and bonding techniques to prepare devices for experimental use.
- Performed **image analysis** on microscopy datasets to quantify cell morphology, proliferation, and responses to experimental stimuli, supporting data-driven interpretation of biological outcomes.

Machine learning research assistant

sep 2024 – Now

University of South Florida

Tampa, USA

- Developed transformer-based and graph neural network models for the Vulcanizing Planning Decision Problem to improve production scheduling accuracy and scalability.
- Built and refined the data-preparation pipeline by generating graph edge files and node feature/label inputs from raw production data, enabling efficient training on structured industrial datasets with over **1 million** data points
- Evaluated and improved model performance using custom loss design, soft-rounding methods, and constraint-aware testing workflows, helping align predictions with demand and resource limits in real-world rubber manufacturing planning.

PROJECTS AND DESIGN TEAMS

Aerostructure Senior Member

Sep 2023 – Jun 2025

University of Toronto Super Mileage – University of Toronto

Toronto, ON

- Collaborated with the aerostructure team with **20+** members to fabricate lightweight vehicle components using carbon-fiber layups, ensuring structural integrity and minimal weight.
- Utilized **SolidWorks** to design steering-wheel prototypes, applying ergonomic and safety criteria to improve driver comfort and interface reliability.
- Executed **CFD** simulations using **Solidworks** for the car to achieve ideal aerodynamic structure and optimize fuel efficiency.
- Operated **3D printers** to fabricate prototype components, adjusting print parameters to ensure dimensional accuracy and reliable part performance.

TECHNICAL SKILLS

Design Software

SolidWorks, Fusion 360, Ansys, KiCad, LT Spice, MS Office

Machining and Power Tools

CNC Mill, Drill Press, Jigsaw, Hand Drill

Programming

Python, Pytorch, Julia, JavaScrip, MATLAB, LaTeX, HTML/CSS